

R U V N E T

Your AI Projects Are Failing.

87% of enterprise AI pilots never reach production.
RuvNet is the infrastructure that fixes that.

500K+

Downloads

12K+

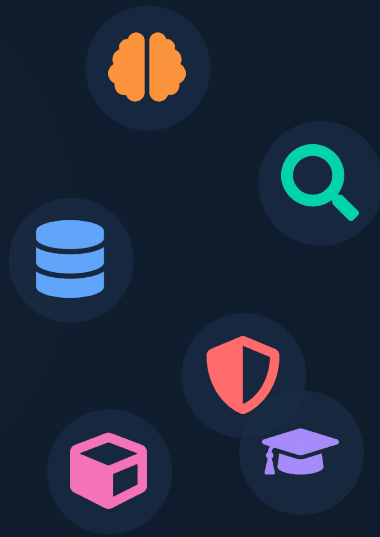
GitHub Stars

100K+

Community

80+

Rust Crates



THE PROBLEM

The \$2.4 Trillion Problem

Enterprise AI spending is massive. But most of it is wasted.

87%

of enterprise AI pilots
never reach production

— Gartner

Why They Fail



No orchestration

Who coordinates 50+ AI agents?



No memory

Agents forget between sessions



No security

Prompt injection undetected



No offline

No internet = no intelligence



No learning

Systems never get smarter

AI vendors give you an engine. You still have to build the car, the road, and the traffic system.

THE GAP

What Every AI Provider Is Missing

They ship the model. You build the other 8 layers.

Capability	OpenAI	Anthropic	Google	RuvNet
Multi-Agent Orchestration	✗	✗	✗	✓
Persistent Memory	✗	✗	✗	✓
Vector Search (<100µs)	✗	✗	✗	✓
Security Middleware	✗	✗	✗	✓
Offline / Air-Gapped	✗	✗	✗	✓
Self-Learning	✗	✗	✗	✓
Anti-Hallucination	✗	✗	✗	✓
WASM Runtime (7.2KB)	✗	✗	✗	✓

Every provider excels at the model layer. None provide the other 8 layers needed for production.

THE STACK

RuvNet: The Complete Agentic Stack

Six integrated modules that turn AI models into production-grade autonomous systems.



Ruflo The Brain

150+ agent types, 5 network topologies

500K downloads



RuVector The Search Engine

61 μ s HNSW search, native PostgreSQL

8,000x faster



AgentDB The Memory

4 memory types, agents remember everything

12,500x retrieval



AIMDS The Immune System

6 attack types in 0.06ms, chaos theory

12K+ req/sec



RVF The Container

AI in a single file, browser or bare metal

5.5KB runtime



SONA The Teacher

System gets smarter with every decision

\$0 per cycle

ARCHITECTURE

How It All Fits Together

Every request flows through a unified pipeline. Security first. Intelligence at every step.



RVF: 3-Tier Compute

WASM (5.5KB) | eBPF (kernel bypass) | Unikernel
(125ms boot)

Nervous System: 5 Bio-Inspired Layers

Sensing (<100ns) | Reflex (<1μs) | Memory | Learning |
Coherence

PostgreSQL Unified Data Plane — 290+ SQL Functions | HNSW Indexing | Drop-in pgvector replacement

Security Envelope — Witness Chains (ML-DSA-65) | TEE Attestations | PII Redaction | DNA Lineage

96 MCP Tools

80+ Rust Crates

49+ npm Packages

25+ ADRs

14 Modules

P R O O F

Real Teams. Real Results.



Fox Flow

Community Builder

12.8M queries/sec

4,000x faster than Redis

Built on RuVector — 12.8 million queries per second.



QE Fleet

Quality Engineering

51 autonomous agents

Across 12 domains

Zero manual coordination across 12 business domains.



Bill Sentry

Healthcare

30 years expertise

Automated billing

30 years of medical AI: first stack that works in production.



Finland Gov

Government

National scale

AI-native transformation

First country-scale agentic deployment.

These aren't demos. These are production deployments.

ONLY WITH RUVNET

Your Data Never Leaves Your Building



How AI Works Today

- ✗ Every query sends data to external servers
- ✗ Trade secrets, customer data transmitted
- ✗ No audit trail of what happens to your data
- ✗ Compliance teams block AI adoption

Result: 87% of AI pilots die in legal review

How RuvNet Works

- ✓ RuVector runs inside YOUR PostgreSQL
- ✓ AIMDS scans every input AND output
- ✓ RVF containers run air-gapped
- ✓ Cryptographic witness chains for audit

Result: HIPAA, GDPR, ITAR, SOC 2 compliant

EXAMPLE: A bank wants AI to analyze 10 years of loan data

OpenAI: Upload to their servers. Compliance blocks it. Project dies.

RuvNet: Deploy on your PostgreSQL. AIMDS enforces PII redaction. Zero data leaves. Compliance approves day one.

ONLY WITH RUVNET

Full AI on Your Phone. No Cloud. No Backend.



RVF cognitive containers shrink an entire AI application into a single file.

Genomic Diagnostics

155 ns/SNP

3 BILLION DNA base pairs in <100ms. On a \$50 phone.

Field Service AI

Zero connectivity

Complete equipment manual + diagnostic AI in a mine shaft.

Military Intelligence

Air-gapped

Classified AI that never touches a network. One file.

Patient Bedside AI

Zero data tx

Drug interactions + vitals on-device. HIPAA trivially solved.

Docker: 50-500 MB | Container: 5-50 MB | RVF WASM: 5.5 KB (9,000x smaller)

BENCHMARKS

Performance That Speaks For Itself



Measured results. Not theoretical. Not marketing projections.

Capability	RuvNet	Industry	Advantage
SWE-Bench Score	84.8%	GPT-4o: 33.2%	2.5x
HNSW Vector Search	61μs	500ms+ (Pinecone)	8,000x
Stream Processing	12.8M QPS	Redis: 3K QPS	4,000x
Threat Detection	0.06ms	Rule-based: 10-50ms	166-833x
Real-Time Learning	<1ms, \$0	Days, \$1K-\$100K	Instant
WASM Runtime	5.5KB	50MB+ containers	9,000x

8,000x

Vector Search

4,000x

Stream
Processing

9,000x

Smaller Runtime

TIMELINE

We Built the Future 8 Months Early

RuvNet doesn't follow the industry roadmap. The industry follows ours.



RuVector

In-process vector search.
Industry still uses external DBs.

AIMDS

Chaos theory security.
Industry still uses rule-based WAFs.

RVF

5.5KB cognitive containers.
Industry still uses Docker.

SONA

Self-optimizing neural arch.
Industry uses static models.

Two standard deviations beyond state of the art. And it's free.

BUSINESS MODEL

This Is Free — And That Should Terrify Your Competitors

Open source. MIT license. Zero licensing cost. The entire 148-capability stack.

MIT

License

\$0

Licensing Cost

148

Capabilities

80+

Rust Crates

WHY FREE MATTERS

Every dollar your competitor spends on licensing, you invest in capability. The gap compounds daily.

The Compound Advantage

Proprietary Training Data

Every interaction generates unique LoRA adapters. Your AI learns from your data.

Institutional AI Memory

Knowledge that no competitor can copy. Your AI knows your company.

Security Threat Intelligence

AIMDS learns from every attack. Attackers train your defense against themselves.

INDUSTRIES

Intelligence at Every Endpoint

The entire vector search engine fits in 7.2 kilobytes. It runs in a browser tab.



Healthcare

HIPAA trivially solved

Diagnostics on-device. Patient data never leaves.



Defense

Classified AI, zero network

Full AI in air-gapped environments.



Field Workers

Full AI without internet

Complete AI on a laptop — offline, anywhere.



Retail

Real-time recommendations

POS AI locally. No cloud round-trip.



IoT / Edge

AI on every device

Every sensor becomes an intelligent endpoint.



Developing World

Same capability as anyone

Rural clinic gets Johns Hopkins-level AI.

This is not a feature. This is a paradigm shift.

DEPLOYMENT

Deploy Anywhere

Every deployment model. SOC 2, HIPAA, GDPR, ITAR compliant.



K8s

Cloud

Kubernetes-native with auto-scaling. Managed infrastructure.



HW

On-Premise

Bare metal deployment. Full data sovereignty. Your hardware.



Hybrid

Cloud + On-Prem

Cloud orchestration with on-premise data. Best of both worlds.



RVF

Air-Gapped

Self-booting cognitive containers. Zero network. Works in a bunker.

BONUS: Browser-Only via WASM

No server. No installation. The 5.5KB WASM runtime runs the full vector search engine in a browser tab.



R U V N E T

Schedule a 30-Minute Architecture Review

See how the 148-capability agentic stack maps to your specific use case.

148+

Capabilities

80+

Rust Crates

500K+

Downloads

MIT

License

Two standard deviations beyond state of the art. Open source. Free. Use it.

ruvnet.com | github.com/ruvnet | hello@ruvnet.com